

Digital Nurture Mockito exercises HandsOn

Exercise 1: Mocking and Stubbing

```
ExternalApi.java MyService.java MyServiceTest.java X
Mockito Hands-On Exercises > MockitoExercises > src > test > java > com > example > MyServiceTest.java > MyServiceTest
1 package com.example;
2
3 import static org.junit.jupiter.api.Assertions.assertEquals;
4 import static org.mockito.Mockito.when;
5
6 import org.junit.jupiter.api.Test;
7 import org.mockito.Mockito;
8
9 public class MyServiceTest {
10
11     @Test
12     void testExternalApi() {
13
14         // Create a mock object
15         ExternalApi mockApi = Mockito.mock(ExternalApi.class);
16
17         // Stub the method
18         when(mockApi.getData()).thenReturn("Mock Data");
19
20         // Inject the mock into the service
21         MyService service = new MyService(mockApi);
22
23         // Call the service method
24         String result = service.fetchData();
25     }
26 }

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises> mvn test
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceTest
WARNING: A Java agent has been loaded dynamically (C:\Users\pasam\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classp
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.830 s -- in com.example.MyServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.557 s
[INFO] Finished at: 2026-07-19T15:25:31+05:30
[INFO] -----
```

Exercise 2: Verifying Interactions

ExternalApi.java

MyService.java

MyServiceTest.java X

Mockito Hands-On Exercises > MockitoExercises > src > test > java > com > example > MyServiceTest.java > MyServiceTest

```
1 package com.example;
2
3 import org.junit.jupiter.api.Test;
4 import org.mockito.Mockito;
5
6 public class MyServiceTest {
7
8     @Test
9     void testVerifyInteraction() {
10
11         // Create a mock object
12         ExternalApi mockApi = Mockito.mock(ExternalApi.class);
13
14         // Inject the mock into the service
15         MyService service = new MyService(mockApi);
16
17         // Call the service method
18         service.fetchData();
19
20         // Verify that getData() was called exactly once
21         Mockito.verify(mockApi).getData();
22     }
23 }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises> mvn test

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceTest
WARNING: A Java agent has been loaded dynamically (C:\Users\pasam\.m2\repository\net\bytebuddy\byte-buddy-agent\1.
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap cla
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.889 s -- in com.example.MyServiceTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 2.972 s
[INFO] Finished at: 2026-07-19T15:35:28+05:30
[INFO] -----
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises>

Exercise 3: Argument Matching

```
ExternalApi.java  MyService.java  MyServiceTest.java X
Mockito Hands-On Exercises > MockitoExercises > src > test > java > com > example > MyServiceTest.java > MyServiceTest

1  package com.example;
2
3  import static org.junit.jupiter.api.Assertions.assertEquals;
4  import static org.mockito.ArgumentMatchers.anyString;
5  import static org.mockito.ArgumentMatchers.eq;
6  import static org.mockito.Mockito.when;
7
8  import org.junit.jupiter.api.Test;
9  import org.mockito.Mockito;
10
11 public class MyServiceTest {
12
13     @Test
14     void testArgumentMatching() {
15
16         // Create mock
17         ExternalApi mockApi = Mockito.mock(ExternalApi.class);
18
19         // Stub for a specific argument
20         when(mockApi.getData(userId: "101")).thenReturn("Employee Data");
21
22         // Inject mock
23         MyService service = new MyService(mockApi);
24     }
25 }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises> mvn test
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceTest
WARNING: A Java agent has been loaded dynamically (C:\Users\pasam\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.12\byte-buddy-agent-1.14.12.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been modified
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.766 s -- in com.example.MyServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.827 s
[INFO] Finished at: 2026-07-19T15:43:31+05:30
[INFO] -----
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises>
```

Exercise 4: Handling Void Methods

```
J ExternalApi.java  J MyService.java  J MyServiceTest.java X
Mockito Hands-On Exercises > MockitoExercises > src > test > java > com > example > J MyServiceTest.java > MyServiceTest
1  package com.example;
2
3  import static org.mockito.Mockito.doNothing;
4  import static org.mockito.Mockito.verify;
5
6  import org.junit.jupiter.api.Test;
7  import org.mockito.Mockito;
8
9  public class MyServiceTest {
10
11     @Test
12     void testVoidMethod() {
13
14         // Create mock
15         ExternalApi mockApi = Mockito.mock(ExternalApi.class);
16
17         // Stub the void method
18         doNothing().when(mockApi).saveData(data: "Mockito");
19
20         // Create service
21         MyService service = new MyService(mockApi);
22
23         // Call the method
24         service.storeData(data: "Mockito");
25
26         // Verify interaction
27         verify(mockApi).saveData(data: "Mockito");
28     }
29 }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises> mvn test
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceTest
WARNING: A Java agent has been loaded dynamically (C:\Users\pasam\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.12)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been renamed
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.760 s -- in com.example.MyServiceTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.799 s
[INFO] Finished at: 2026-07-19T15:50:19+05:30
[INFO] -----
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises>
```

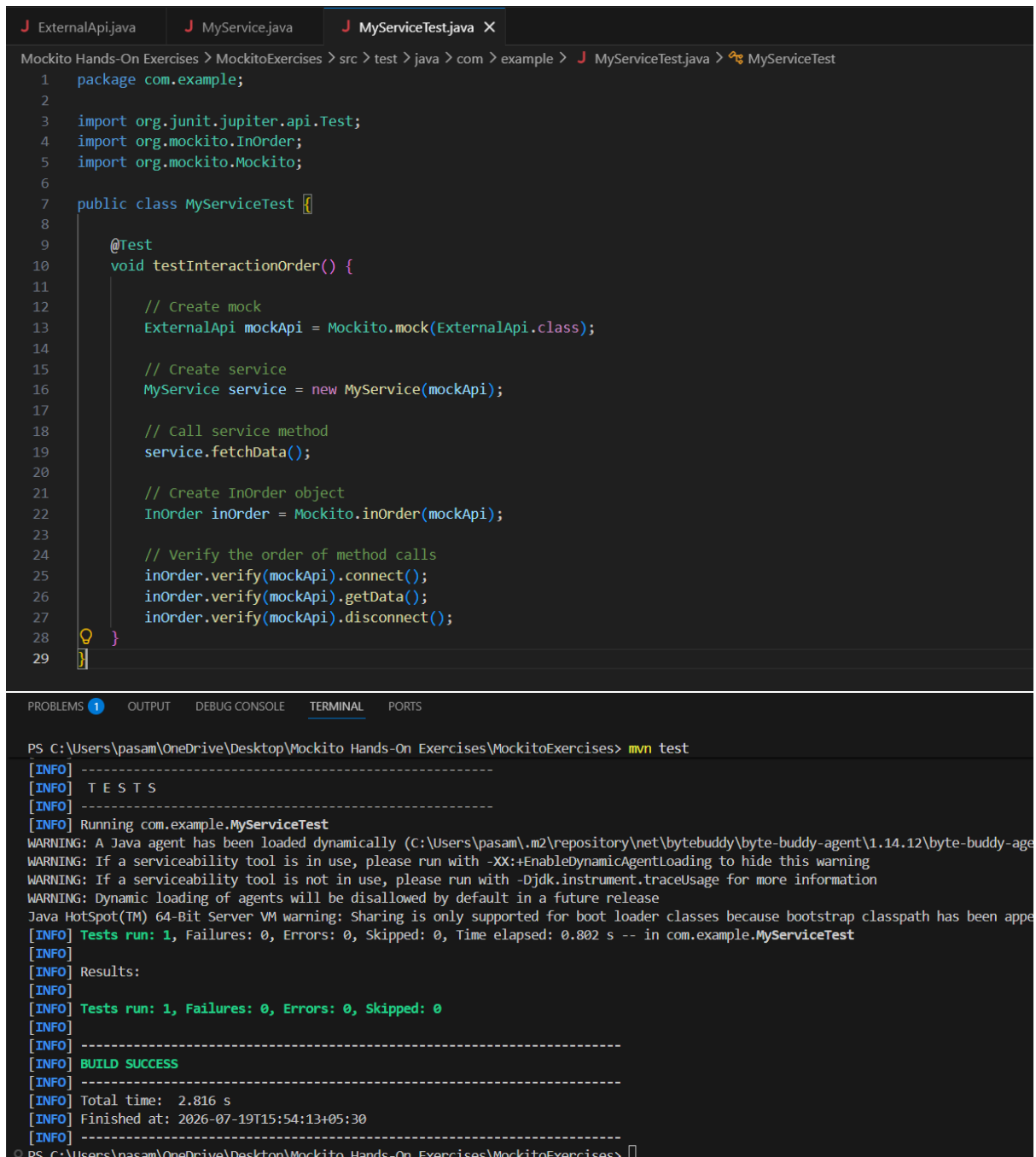
Exercise 5: Mocking and Stubbing with Multiple Returns

```
J ExternalApi.java  J MyService.java  J MyServiceTest.java X
Mockito Hands-On Exercises > MockitoExercises > src > test > java > com > example > J MyServiceTest.java > MyServiceTest

1  package com.example;
2
3  import static org.junit.jupiter.api.Assertions.assertEquals;
4  import static org.mockito.Mockito.times;
5  import static org.mockito.Mockito.verify;
6  import static org.mockito.Mockito.when;
7
8  import org.junit.jupiter.api.Test;
9  import org.mockito.Mockito;
10
11 public class MyServiceTest {
12
13     @Test
14     void testMultipleReturns() {
15
16         // Create mock
17         ExternalApi mockApi = Mockito.mock(ExternalApi.class);
18
19         // Return different values on consecutive calls
20         when(mockApi.getData())
21             .thenReturn("First Call")
22             .thenReturn("Second Call")
23             .thenReturn("Third Call");
24
25     }
26 }

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises> mvn test
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceTest
WARNING: A Java agent has been loaded dynamically (C:\Users\pasam\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.12\byte-buddy-agent-1.14.12.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been modified
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.760 s -- in com.example.MyServiceTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 2.857 s
[INFO] Finished at: 2026-07-19T15:52:16+05:30
[INFO] -----
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises>
```

Exercise 6: Verifying Interaction Order



The screenshot shows an IDE with three tabs: `ExternalApi.java`, `MyService.java`, and `MyServiceTest.java`. The `MyServiceTest.java` tab is active, displaying the following code:

```
1 package com.example;
2
3 import org.junit.jupiter.api.Test;
4 import org.mockito.InOrder;
5 import org.mockito.Mockito;
6
7 public class MyServiceTest {
8
9     @Test
10    void testInteractionOrder() {
11
12        // Create mock
13        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
14
15        // Create service
16        MyService service = new MyService(mockApi);
17
18        // Call service method
19        service.fetchData();
20
21        // Create InOrder object
22        InOrder inOrder = Mockito.inOrder(mockApi);
23
24        // Verify the order of method calls
25        inOrder.verify(mockApi).connect();
26        inOrder.verify(mockApi).getData();
27        inOrder.verify(mockApi).disconnect();
28    }
29 }
```

Below the code editor, the `TERMINAL` tab is active, showing the output of the command `mvn test`:

```
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises> mvn test
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceTest
WARNING: A Java agent has been loaded dynamically (C:\Users\pasam\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.12\byte-buddy-agent.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.882 s -- in com.example.MyServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.816 s
[INFO] Finished at: 2026-07-19T15:54:13+05:30
[INFO] -----
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises>
```

Exercise 7: Handling Void Methods with Exceptions

```
J ExternalApi.java J MyService.java J MyServiceTest.java X
Mockito Hands-On Exercises > MockitoExercises > src > test > java > com > example > J MyServiceTest.java > MyServiceTest
1 package com.example;
2
3 import static org.junit.jupiter.api.Assertions.assertThrows;
4 import static org.mockito.Mockito.doThrow;
5 import static org.mockito.Mockito.verify;
6
7 import org.junit.jupiter.api.Test;
8 import org.mockito.Mockito;
9
10 public class MyServiceTest {
11
12     @Test
13     void testVoidMethodThrowsException() {
14
15         // Create mock
16         ExternalApi mockApi = Mockito.mock(ExternalApi.class);
17
18         // Stub the void method to throw an exception
19         doThrow(new RuntimeException("Database Error"))
20             .when(mockApi)
21             .saveData(data: "Mockito");
22
23         // Create service
24         MyService service = new MyService(mockApi);
25
26     }
27 }

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises> mvn test
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.MyServiceTest
WARNING: A Java agent has been loaded dynamically (C:\Users\pasam\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.12\byte-buddy-agent-1.14.12.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.080 s -- in com.example.MyServiceTest
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.218 s
[INFO] Finished at: 2026-07-19T15:56:51+05:30
[INFO] -----
PS C:\Users\pasam\OneDrive\Desktop\Mockito Hands-On Exercises\MockitoExercises>
```